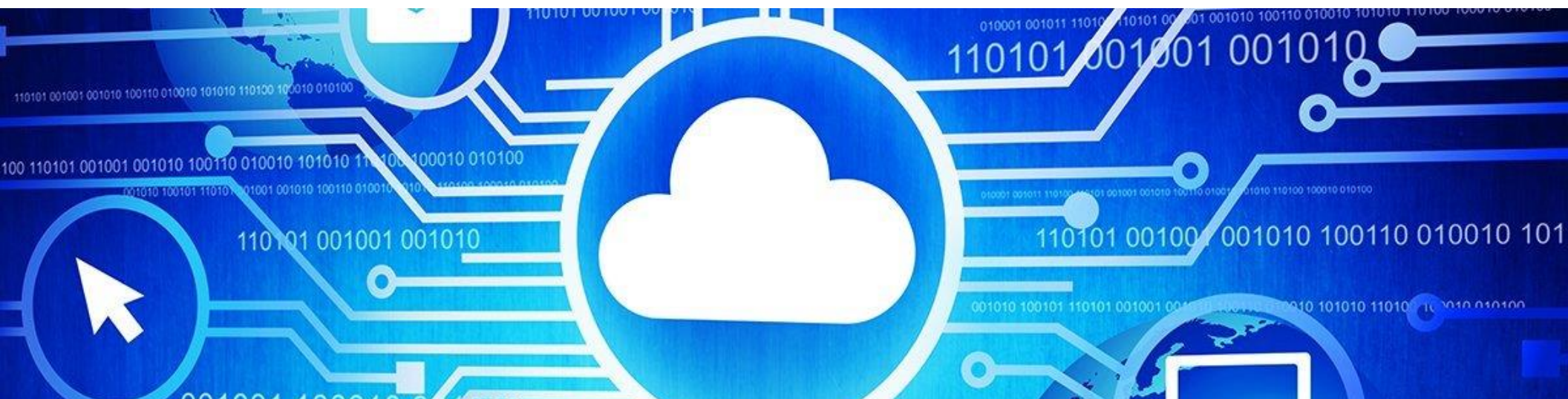


Cloud Backup: What You Need to Know to Get Started



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In this e-guide:

Implementing cloud data backup can help bolster an organization's data protection strategy without increasing the workload of IT staff.

However, there are many intricate parts to even the simplest cloud backup strategies that IT staff should be aware of to fully reap all of its benefits.

This cloud backup guide can help get you started. Explore:

- How cloud backup works
- Types of backup
- Pros vs. cons of cloud backup
- Cloud backup vs. cloud DR
- Cloud vs. local back up
- Security considerations
- Acronis vs. Druva vs. 8 other vendors

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■ What is Cloud Backup and How Does it Work?

Margaret Rouse, [WhatIs.com](#)

Cloud backup, also known as online backup or remote backup, is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure or catastrophe. The secondary server and storage systems are usually hosted by a third-party service provider, who charges the backup customer a fee based on storage space or capacity used, data transmission bandwidth, number of users, number of servers or number of times data is accessed.

Implementing cloud data backup can help bolster an organization's data protection strategy without increasing the workload of information technology (IT) staff. The labor-saving benefit may be significant and enough of a consideration to offset some of the additional costs associated with cloud backup, such as data transmission charges.

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Most [cloud subscriptions](#) run on a monthly or yearly basis. While initially used mainly by consumers and home offices, online backup services are now commonly used by small and medium-sized businesses (SMBs) as well as larger enterprises to back up some forms of data. For larger companies, cloud data backup may serve as a supplementary form of backup.

What is the cloud?

[Cloud computing](#) is a general term that refers to hosted services that are delivered over the internet. Different from traditional web hosting, the services on the cloud are sold on demand, are offered in an elastic manner - meaning the customer can use as much or as little of the service as needed -- and are managed completely by the service provider. Additionally, a cloud can be private or public. A [public cloud](#) sells services to anyone on the internet, such as how Amazon Web Services ([AWS](#)) operates, while a [private cloud](#) supplies hosted services to a limited number of users.

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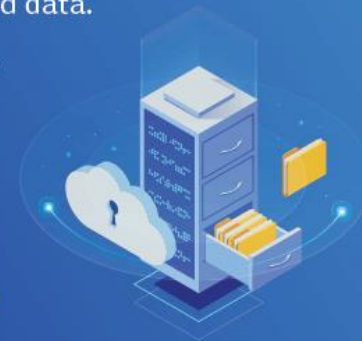
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How cloud backup works

In an organization's data center, a backup application copies data and stores it on a different media or another storage system for easy access in the event of a recovery situation. While there are multiple options and [approaches to off-site backup](#), cloud backup serves as the off-site facility for many organizations. In an enterprise, the company might own the off-site server if it hosts its own cloud service, but the chargeback method would be similar if the company uses a service provider to manage the cloud backup environment.

Basic steps of cloud backup

- Files are designated for backup, typically new or updated data.
- The backed-up files are encrypted and sent off site to the cloud.
- Organizations can choose between public cloud, private cloud or cloud-to-cloud (C2C) backup methods.
- Files can be easily accessed and readily available in the event of a recovery.



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There are a variety of [approaches to cloud backup](#), with available services that can easily fit into an organization's existing data protection process.

Varieties of cloud backup include:

- **Backing up directly to the public cloud.** One way to store organizational resources is by [duplicating resources in the public cloud](#). This method entails writing data directly to cloud providers, such as AWS or Microsoft Azure. The organization uses its own backup software to create the data copy to send to the cloud storage service. The cloud storage service then provides the destination and safekeeping for the data, but it does not specifically provide a backup application. In this scenario, it is important that the backup software is capable of interfacing with the cloud's storage service. Additionally, with public cloud options, IT professionals may need to look into supplemental data protection procedures.
- **Backing up to a service provider.** In this scenario, an organization writes data to a cloud service provider that offers backup services in a managed data center. The backup software that the company uses to send its data to the service may be provided as part of the service, or the service may support specific commercially-available backup applications.
- **Choosing a cloud-to-cloud (C2C) backup.** These services are among the newest offerings in the cloud backup arena. They specialize in backing up data that already lives in the cloud, either as data created using a software as a service ([SaaS](#)) application or as data stored in a cloud backup service. As its name suggests, a [cloud-to-cloud backup service](#) copies data from one cloud to another cloud. The cloud-to-

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cloud backup service typically hosts the software that handles this process.

- **Using online cloud backup systems.** There are also hardware alternatives that facilitate backing up data to a cloud backup service. These appliances are all-in-one backup machines that include backup software and disk capacity along with the backup server. The appliances are about as close to plug-and-play as backup gets, and most of them also provide a seamless (or nearly so) link to one or more cloud backup services or cloud providers. The list of vendors that offer backup appliances that include cloud interfaces is long, with Quantum, Unitrends, Arcserve, Rubrik, Cohesity, Dell EMC, StorageCraft and Asigra active in this arena. These appliances typically retain the most recent backup locally, in addition to shipping it to the cloud backup provider, so that any required recoveries can be made from the local backup copy, saving time and transmission costs.

When an organization engages a cloud backup service, the first step is to complete a [full backup](#) of the data that needs to be protected. This initial backup can sometimes take days to finish uploading over a network as a result of the large volume of data that is being transferred. In a [3-2-1 backup strategy](#), where an organization has three copies of data on two different media, at least one copy of the backed up data should be sent to an off-site backup facility so that it is accessible even if on-site systems are unavailable.

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Using a technique called [cloud seeding](#), a cloud backup vendor sends a storage device -- such as a disk drive or tape cartridge -- to its new customer, which then backs up the data locally onto the device and returns it to the provider. This process removes the need to send the initial data over the network to the backup provider.

If the amount of data in the initial backup is substantial, the cloud backup service may provide a full storage array for the seeding process. These arrays are typically small [network-attached storage](#) (NAS) devices that can be shipped back and forth relatively easily. After the initial seeding, only changed data is backed up over the network.

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Cloud backup checklist: Key features

FEATURE	REASON FOR RELEVANCE
Encryption of data in transit and at rest in the cloud	Prevents access to the data by cloud service provider staff and other unauthorized users
SAS 70/SSAE 16 Type II compliance of the cloud service provider	Ensures the service provider has strong IT controls in place; a must-have for public companies and for industries with additional regulatory requirements
Data deduplication	Reduces storage cost; source-side deduplication also reduces bandwidth requirements
Hybrid cloud backup option	Cached on-premises backups eliminate long restore times of a pure cloud backup offering
Getting data in and out of the cloud via physical device	Options of initial "seeding" of backups and delivery of restore data via physical disk to minimize initial backup and restore times
Incremental forever	Ensures efficient ongoing backups
Sub-file-level backup of changed files	Minimizes the amount of data to be backed up, especially with large files
Bandwidth features	Throttling and scheduling to minimize the impact of backups on other users

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How data is restored

[Cloud backup services](#) are typically built around a client software application that runs on a schedule determined by the purchased level of service and the customer's requirements. For example, if the customer has contracted for daily backups, the application collects, compresses, encrypts and transfers data to the cloud service provider's servers every 24 hours. To reduce the amount of bandwidth consumed and the time it takes to transfer files, the service provider might only provide [incremental backups](#) after the initial full backup.

Cloud backup services often include the software and hardware necessary to protect an organization's data, including applications for Exchange and SQL Server. Whether a customer uses its own backup application or the software the cloud backup service provides, the organization uses that same application to restore backed up data. Restorations could be on a file-by-file basis, by volume or a full restoration of the complete backup.

If the volume of data to be restored is very large, the cloud backup service may ship the data on a complete storage array that the customer can hook up to its servers to recover its data. This is, in effect, a reverse seeding process. Restoring a large amount of data over a network can take a long time.

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A key feature of cloud backup restorations is that they can be done anywhere from nearly any kind of computer. For example, an organization could recover its data directly to a disaster recovery site in a different location if its data center is unavailable.

Types of backup

In addition to the various approaches to cloud backup, there are also [multiple backup methods to consider](#). While cloud backup providers give customers the option to choose the backup method that best fits their needs and applications, it is important to understand the [differences among the three main types](#).

1. Full backups copy the entire data set every time a backup is initiated. As a result, they provide the highest level of protection. However, most organizations cannot perform full backups frequently because they can be time-consuming and take up too much storage capacity.
2. Incremental backups only back up the data that has been changed or updated since the last backup. This method saves time and storage space, but can make it more difficult to perform a complete restore. Incremental is a common form of cloud backup because it tends to use fewer resources.

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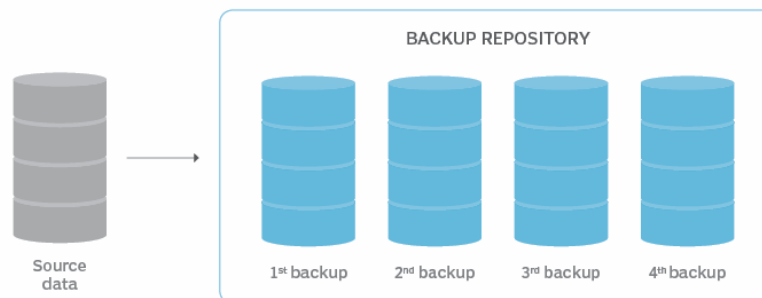
3. **Differential backups** are similar to incremental backups because they only contain data that has been altered. However, differential backups back up data that has changed since the last *full* backup, rather than the last backup in general. This method solves the problem of difficult restores that can arise with incremental backups.

➤ Click on image to view full size

Full vs. incremental vs. differential backup

Full backup

Data is copied in its entirety every time.



Incremental backup

Data is copied in its entirety to begin with, and then only new or updated data is backed up each time a backup is initiated after that.

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Pros and cons

Before choosing cloud backup as a backup strategy, it is important to [weigh the potential pros and cons](#) that are associated with using a third-party to store data. The advantages of cloud backup include:

- Generally, it is cheaper to back up data using a cloud backup service compared to building and maintaining an in-house backup operation. The associated cloud backup costs will rise as the volume of backup data rises, but the economies are likely to continue to make cloud backup an attractive choice. Some providers may offer free cloud backup, but the amount of backup capacity is typically limited which makes free backup appropriate for some home users and only the smallest of companies.
- The cloud is scalable, so even as a company's data grows, it can still be easily backed up to a cloud backup service. But organizations need to be wary of escalating costs as data volume grows. By weeding out dormant data and sending it to an archive, a company can better manage the amount of data it backs up to the cloud.
- Managing cloud backups is simpler because service providers take care of many of the management tasks that are required with other forms of backup.
- Backups are generally more secure against [ransomware](#) attacks because they are performed outside of the office network. Backup data is typically encrypted before it is transmitted from the customer's

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site to the cloud backup service, and usually remains encrypted on the service's storage systems.

- Cloud backups help lower the risk of [common data backup failures](#) caused by improper storage, physical media damage or accidental overwrites.
- A cloud backup service can help to consolidate a company's backup data because the service can back up main data center storage systems, remote office servers and storage devices, and end-user devices such as laptops and tablets.
- Backed up data is accessible from anywhere.

Despite its many benefits, there are some disadvantages and [challenges to using a cloud backup service](#), including:

- The backup speed depends on bandwidth and latency. For example, when many organizations are using the internet connection, the backup could be slower. This could be bothersome when backing data up, but could be an even greater issue when it is necessary to recover data from the service.
- Costs can escalate when backing up large amounts of data to the cloud.
- As with [any use of cloud storage](#), data is moved outside of an organization's buildings and equipment and into the control of an outside provider. Therefore, it is incumbent to learn as much as possible about the cloud backup provider's equipment, physical security procedures, data protection process and fiscal viability.

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Best practices

While strategies, technologies and providers widely vary, there are [several agreed upon best practices](#) when it comes to implementing cloud backup in the enterprise. In general, a few guidelines are:

- Understand all aspects of the cloud backup provider [service-level agreement](#) (SLA) such as how data is backed up and protected, where vendor offices are located and how costs accumulate over time.
- Do not rely on any one method or storage medium for backup.
- Test [backup strategies](#) and data recovery checklists to ensure they are sufficient in the case of a disaster.
- Have administrators routinely monitor cloud backups to make sure processes are successful and uncorrupted.
- Choose a data restore destination that is easily accessible and does not overwrite existing data.
- Make decisions about specific data or files to back up based on the criticality of the information to business operations.
- Use metadata properly to enable the quick location and restoration of specific files.
- Consider using encryption for data that must stay confidential.

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Special considerations

When [choosing a cloud backup service provider](#), there are a few additional considerations to weigh. Some companies have special needs related to data protection, but not all cloud backup providers are able to meet those needs. For example, if a company must comply with a specific regulation such as [HIPAA](#) or [GDPR](#), the cloud backup service needs to be certified as compliant with data handling procedures as defined by that regulation. While an outside firm provides the backup, the customer is still responsible for the data, and could face serious consequences -- including steep fines -- if the cloud backup provider does not maintain the data appropriately.

[Data archiving](#) is another special consideration when selecting a cloud backup service. Archiving is different from routine data backup. Archived data is data that is not currently needed but still needs to be retained. Ideally, that data should be removed from the daily backup stream because it is likely unchanged and it unnecessarily increases the volume of backup data transmissions. Some cloud backup providers offer archiving services to complement their backup products. Archive data is generally stored on equipment geared to longer retentions and infrequent access, such as tape or low-performing disk systems. That type of storage is generally less expensive than storage used for active backups.

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Cloud backup vs. cloud DR

Cloud backup and [cloud disaster recovery](#) are not the same, but they are connected. While cloud backup services can be tapped to recover data and resume operations after a disruptive event, it should be noted that they are not necessarily specifically oriented to provide all the features and services that a true [DRaaS](#) offering would provide.

For example, to use the data that was saved to a cloud backup service to recover from a disaster, it would have to include much more than just data files, such as operating systems, application software, drivers and utilities. Users would have to set up their backup routines to include those elements specifically, such as by mirroring entire servers to the cloud backup service.

More importantly, a true DRaaS not only has the data and system and application software ready to be accessed, it also provides the necessary servers -- physical or virtual -- and storage resources to spin up their clients' servers and applications so that they continue to operate and carry on with their business.

An organization must consider if the disaster recovery provider has enough bandwidth and resources to handle the data transfer, and thus how long it will take to perform a recovery. Testing is important and often easier than

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with traditional disaster recovery, because many providers offer automated tests.

A cloud backup provider may also offer disaster recovery in the cloud. Cloud disaster recovery is particularly attractive for smaller businesses that do not have the funding or resources to support their own DR site. The cloud data center should be far enough away from the organization using it to ensure recovery from any disaster.

Cloud vs. local backup

When looking into data backup options, two main product categories are [cloud backup and local backup](#). Local backup, also known as traditional backup, is the process of storing a copy of data on-site at the organization. In this approach, backup software is used to manage and copy the data to hardware such as tapes, disks or network-attached storage devices.

In the enterprise, cloud data backup services were initially used for noncritical data. Traditional backup was considered better for critical data that requires a short recovery time objective ([RTO](#)) because there are physical limits as to how much data can be moved in a given amount of time over a network. When a large amount of data needs to be recovered, it may need to be shipped on tape or some other portable storage media. However, with most cloud backup schemes -- whether controlled by a user's backup

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software, the cloud backup service app or a backup appliance -- the most recently backed up data is retained on site as well as spun off to the cloud service. This arrangement provides all the benefits of cloud backup while leaving a local copy of the data that can be used for quick recoveries.

Tape backup requires data to be copied from a primary storage device to a tape cartridge. Cartridges have grown dramatically in capacity in recent years. **LTO-8** tapes, released in late 2017, can store 12 TB of uncompressed data and 30 TB compressed. Tape is a strong storage medium in an age of exponential data growth. In addition to their capacity benefits, tapes are comparatively inexpensive to own and operate. However, the restore process can be slow because access is sequential.

While the cloud appears to offer unlimited capacity, costs rise dramatically depending on how much storage an organization needs. Although access is not sequential like with tape, restore times still depend on the internet or private communications lines, and require an appropriate amount of bandwidth. Cloud service providers take some of the backup management work out of the process for organizations. The process of backing up to tape and maintaining the cartridges is essentially up to the organization. There is more flexibility in the process of restoring from cloud backup because an organization can restore to several different devices, including laptops and phones.

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Cloud and tape both provide protection from cyberattacks, such as ransomware. Cloud backups are useful in the event of an attack because they are off-site. Tape backups are even more secure because they are offline.

[Disk](#), while not as portable as tape, is another common medium for backup. The biggest benefit to a disk is access speed. Disks offer random access and often top cloud and tape for restore speed. Disk-based backups are typically performed continuously throughout the day, while tape backs up less regularly. A disk-based backup is self-contained, with less personnel interaction than with tape, making the risk of human error smaller. Disk-based backups can be expensive, often costlier than tape or cloud. The life span of disk is shorter than tape, and its durability is weaker than that of tape. As long as the service provider is still in business, the life span of a cloud backup could be longer than that of disk or tape.

NAS backup uses a type of appliance that connects to a network, rather than a desktop or server, to accomplish a local backup. These appliances allow multiple devices and users on the same network to store, access and share data wirelessly. Both [NAS and cloud backup](#) offer strong data protection, high security and efficient recovery times, but since NAS appliances are located on the same local area network ([LAN](#)) as the devices being backed up, NAS backups are quicker to perform than cloud backups.

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However, cloud backups can provide a lower initial cost and a higher reliability in the case of on-site disasters.

The following chart helps to illustrate when cloud data backups should be considered as a viable option.

When should you use cloud backups?		
BACKUP FACTOR	CLOUD STORAGE	TRADITIONAL BACKUP
Amount of data	Best when the total amount to protect is less than 100 GB per 1 Mb of network bandwidth. For example, 100 GB can be supported by a 1 Mb WAN connection (such as DSL).	For large amounts of data or for environments with limited network connectivity, traditional backup techniques are more appropriate.
Rate of change	Best when the rate of change is less than 10% of the total data per month.	For data that changes frequently, traditional backup methods that use local disk and tape, with tape transport off-site, are more appropriate.

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With a proper retention policy, cloud backups can reduce or even replace the need for off-site tape storage, so organizations are making the switch from disk-to-disk-to-tape ([D2D2T](#)) strategies to disk-to-disk-to-cloud ([D2D2C](#)).

Flexibility is another benefit of the cloud because no additional hardware is needed.

File sync and share

There is often confusion among the definitions of cloud backup, cloud storage and cloud sync, often referred to as file sync and share (FSS). There are similarities among the three, but [they are different processes](#).

File sync and share services allow users to create folders online where they can store and access files stored on personal computers and servers. As the name implies, these services can automatically update files to their latest versions whether online or stored locally. They also make it easy to share files with colleagues or clients, and to work collaboratively. Cloud sync providers include Box, Dropbox, Google Drive and Microsoft OneDrive.

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Some companies rely on file sync and share services to back up their data as well. While this approach might be acceptable for a small amount of data, it is not appropriate for large data volumes or a company's critical data as these services tend to lack the types of management and control features the cloud backup offers. Also, given their user-oriented approach to data handling, data may become vulnerable if mishandled by sync and share participants.

Hybrid cloud backup

[Hybrid cloud backup](#) providers connect traditional backups to the cloud. This strategy is useful for organizations that produce a large volume of data and need quick restore access. With one approach, a network-attached storage ([NAS](#)) appliance serves as a local backup target and syncs backup data to the cloud. When an organization needs a quick restore, the data is available in the on-site NAS. If an organization loses its primary site, the cloud backup is still available. This method can also be referred to as D2D2C backup. In another hybrid approach, an organization uses both the [public and private cloud for backup](#).

It is difficult to get data consistency with hybrid cloud backup, especially if the data transfer takes a long time. [Point-in-time snapshots](#) and continuous backups help, but costs rise as backup frequency increases.

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Comparatively, in a pure cloud backup scenario, backups go directly to the service provider's cloud.

Cost

Third-party cloud backup initially gained popularity with SMBs and home users because of its convenience. Today, cloud backup services have become more sophisticated and can offer the same level of data protection, if not greater, as in-house data backup.

The technology has an initial upfront cost to implement, but its lower monthly or yearly payment plans appeal to many smaller operations. Capital expenditures for additional hardware are not required and backups can be run dark. However, the cost of keeping data in the cloud for years does add up. In addition, costs rise as the amount of data backed up to the cloud increases. A system of effective data grooming and archiving can help keep costs down.

In terms of return on investment, it is important for an organization to consider the long-term costs of backing up to the cloud. A five-year projection is recommended to properly estimate future expenses and to decide whether the cloud will help an organization break even after initial costs. After these costs are offset, ROI on cloud-based backups can be determined.

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Pricing models vary by vendor, but it is important to look out for hidden costs in cloud backup services. While most products for backing up to the cloud are sold using a price-per-gigabyte-per-month payment model, providers can also use a sliding scale model, set usage minimums and add transaction costs.

Security

Security is an important element in the cloud backup process. The three main considerations are often referred to as the security CIA: confidentiality, integrity and availability.

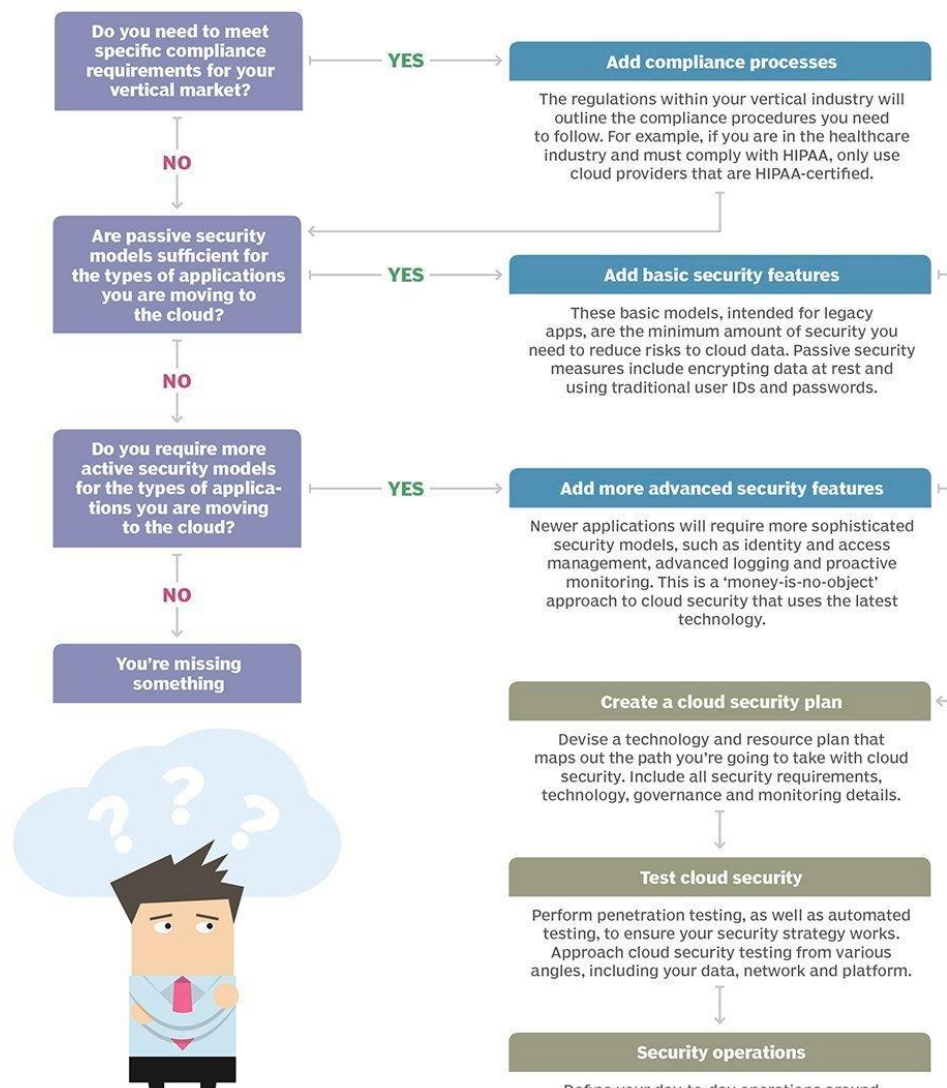
Most data will move across the public internet on its way to the cloud, so for confidentiality, many cloud backup providers encrypt data throughout the process: at the original location, during transit and at rest in the provider's data center. A user or the provider holds the [encryption key](#). Most organizations prefer to hold their encryption keys, and providers should offer this option. Types of network encryption include Secure Sockets Layer (SSL) and Transport Layer Security protocols.

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For integrity, users must determine if data is the same when it is read back or if it was corrupted. [Object storage](#) offers built-in integrity checks.

Availability takes the restoration process into account. It asks: Will data be available in a timely fashion in a disaster recovery (DR) situation?

Access control is also important. An organization tightens security by limiting access to cloud backups. Furthermore, write-once, read-only access protects backup data from being overwritten, altered or deleted.

Vendor options

Approaches to online backup services vary, so an organization should take a close look at SLAs, pricing plans and long-term costs before choosing a provider. Examples of cloud data backup vendor options include the following:

- **Acronis** offers Backup Cloud, a hybrid cloud backup-as-a-service product. Acronis Backup Cloud protects virtual, physical and cloud environments, and includes a pay-as-you-go business model.
- **Arcserve** expanded its Unified Data Protection (UDP) offering with its purchase of Zetta. The product includes Arcserve UDP Cloud Direct direct-to-cloud disaster recovery and backup. The cloud protection is targeted at the midmarket.

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- **Asigra** is a cloud backup pioneer. Its Cloud Backup features embedded malware engines to prevent ransomware from getting into backups.
- **Backblaze** offers personal and business cloud backup, as well as cloud storage. Backblaze stores data on its open source Storage Pods hardware platform and cloud-based Backblaze Vault file system. Backup data through Backblaze is accessible through a web browser on mobile devices and computers. Restores are downloaded over SSLs.
- **Carbonite** sells to consumers, SMBs and enterprises. The company's offerings back up documents, email, music, photos and settings and are available for Windows and Mac users. In March 2018, Carbonite acquired rival Mozy from Dell EMC and incorporated its services into its offerings.
- **Druva** cloud backup features three major offerings. The enterprise-level Druva inSync is targeted at endpoints and backs up data across physical and public cloud storage, while Phoenix is a software agent used to back up and restore data sets in the cloud for distributed physical and virtual servers. In addition, in 2018, Druva acquired CloudRanger for AWS data protection.
- **IDrive** cloud backup is geared toward consumers and small businesses and includes snapshots, a syncing service and hybrid data protection.
- **Microsoft Azure Backup** automatically sends backups to the Azure cloud. Azure Site Recovery automates replication to back up private Windows infrastructure.

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- **Unitrends** allows customers to back up indefinitely to its private cloud with Forever Cloud. It offers several disaster recovery as a service (DRaaS) options for recovery.
- **Veeam Software** provides cloud backup through its Cloud Connect product. Service providers can partner with Veeam to create a backup and recovery target in the cloud.

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SearchDataBackup.com offers IT professionals – charged with building and managing efficient backup and restore policies and processes – relevant, independent advice, expert tips, and industry news. And because of ever-expanding data, the rise of server virtualization, and the development of emerging technologies, it is more important than ever for storage pros to implement an effective backup and data protection strategy.

Whether you're building your data protection plan or you're looking for guidance on the deployment and use of today's backup and restore technologies – such as data deduplication, continuous data protection (CDP), archiving, tape and disk options, snapshots, and more – SearchDataBackup.com has you covered. Welcome.

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